

A Genetic Algorithm (GA) Approach to Portfolio Design Based on Market Movements and Asset Valuations

Summary

This expanded briefing distills the objectives, theoretical underpinnings, experimental design, empirical findings, and practical implications of the IEEE Access article titled “A Genetic Algorithm (GA) Approach to Portfolio Design Based on Market Movements and Asset Valuations.” The paper introduces CAPM+, an investment framework that integrates risk-adjusted momentum with Capital Asset Pricing Model (CAPM) mis-pricing signals. A binary-encoded GA identifies a subset of stocks that jointly maximise Sharpe ratio and aggregate undervaluation. Ten-year back-tests across South-Korean and U.S. large-cap universes reveal cumulative returns exceeding 400 %, superior drawdown control, and favourable risk metrics relative to both momentum-only portfolios and their benchmark indices. The authors thus demonstrate that combining valuation heuristics with technical momentum inside a GA search yields robust, geographically portable portfolios.

1. Background and Motivation

Classic mean-variance optimisation (Markowitz, 1952) assumes normally distributed returns and frictionless trading—assumptions seldom met in practice. Moreover, its quadratic programming grows intractable for hundreds of assets. Heuristic meta-heuristics such as GAs provide dimensionality relief, yet they often rely purely on past returns or variance to shape fitness, overlooking valuation. Concurrently, momentum anomalies (Jegadeesh & Titman, 1993) offer persistent excess returns, but their susceptibility to regime shifts can erode performance. CAPM+, conceived by Lim, Kim, and Ahn (2020), fuses these streams: momentum gauges trend persistence while CAPM mis-pricing estimates fundamental upside. The strategy’s guiding hypothesis states that a diversified basket of fundamentally undervalued momentum winners will outperform broader equity indices on a risk-adjusted basis.

2.Methodology

2.1 Data Sets and Pre-Processing

-KOSPI-200: 190 continuously listed constituents (Jan 2008 – Dec 2018).

-S&P 500: 464 constituents with uninterrupted price histories over the same horizon.

Daily data are aggregated to monthly returns; survivorship bias is mitigated by including delisted stocks. Bid-ask spread and liquidity are approximated via a 0.015 % commission plus 0.3 % transaction tax on each trade.

2.2 Signal Engineering

Momentum Signal : Monthly Sharpe ratio calculated over 1, 3, and 6 month rolling windows. The shorter windows capture rapid market rotations whereas longer windows smooth transient noise.

Valuation Signal : A dynamic beta (β) is computed through a 36 month rolling OLS regression of stock returns on the market index. CAPM's security market line then yields an expected return; the spread between actual realised return and CAPM expectation serves as an undervaluation proxy (positive if underpriced).

Both signals are Z-standardised to comparable scales before combination.

2.3 Genetic Algorithm Design

Encoding & Population : Chromosome length equals the number of stocks (190 or 464). An initial population of 300 chromosomes is seeded via uniform random inclusion probability of 0.15 to avoid over-concentration.

Fitness Function : $F = w_1 \cdot \text{Sharpe} + w_2 \cdot \text{Aggregate Undervaluation} - \lambda \cdot \text{Turnover}$, where λ penalises excessive rebalancing. Weights (w_1, w_2) are tuned via cross-validated grid search (best: 0.7, 0.3). Turnover penalty preserves net returns after friction.

Operators : Tournament selection (size = 5) chooses parents; two-point crossover at 100 % probability promotes diversity; 1 % per-gene mutation flips inclusion bits; elitism clones the top 5 % chromosomes each generation to guard against performance regressions. Convergence is usually attained by generation 150 (KOSPI) or 300 (S&P 500).

2.4 Rebalancing Protocol

The GA is run at every month end. Selected stocks are equal-weighted and held for the look-back window length. Capital released from selling exclusions is redeployed

in newly admitted inclusions. The rolling process produces 120 overlapping portfolios across the ten year horizon.

3.Results and Comparative Analysis

3.1 Headline Performance Metrics

- KOSPI-200 CAPM+ (1 month window): 432 % cumulative return, annualised Sharpe 1.23, max drawdown –18 %.
- KOSPI Index: 156 % cumulative return, Sharpe 0.55, max drawdown –41 %.
- S&P 500 CAPM+ (1 month window): 407 % cumulative return, Sharpe 1.16, max drawdown –15 %.
- S&P 500 Index: 239 % cumulative return, Sharpe 0.78, max drawdown –34 %.

3.2 Sensitivity Tests

Robustness checks varied momentum horizon, fitness weights, and transaction cost assumptions. Cumulative returns remained $> 2.5\times$ their benchmarks in 84 % of parameter combinations, highlighting strategy resilience. Stress tests under 2008 and 2011 crisis windows show CAPM+ suffering shallower drawdowns and recovering to new highs within 8 months, versus 22 months for the indices.

3.3 Why Does it Work?

Undervaluation acts as a valuation anchor, trimming exposure to overheated momentum names susceptible to reversals. Conversely, the momentum filter sidesteps classic value traps. The GA's binary selection scheme implicitly diversifies across sectors without an explicit sector constraint, thereby mitigating idiosyncratic risk. By tackling the portfolio as a combinatorial subset problem, the GA avoids estimating a noisy full covariance matrix, sidestepping the curse of dimensionality endemic to Markowitz optimisers.

4.Implementation Considerations and Limitations

Practical Adoption : Institutional deployment would require intraday liquidity screens to ensure fill quality on monthly turnover. Equal-weight allocation simplifies execution but could be extended to risk parity or inverse volatility schemes without altering GA selection mechanics.

Limitations :

- CAPM beta assumes linearity with market returns, ignoring multi-factor effects
- Genetic meta-heuristics can converge to local optima—multiple random restarts are advised
- Survivorship-biased universe culling was mitigated but not eliminated for the S&P 50
- The study excludes dividends, potentially understating benchmark returns
- Transaction costs are South-Korea specific—U.S. brokers might impose different fee schedules

5.Conclusion and Future Research Directions

The study provides compelling evidence that weaving valuation metrics into momentum-driven GA optimisation can deliver durable alpha in disparate equity markets. Next-generation research could investigate:

- Adding macro risk regimes into the fitness function
- Hybridising GA with deep reinforcement learning for adaptive reweighting
- Extending the universe to small-caps and emerging markets
- Evaluating ESG-tilted CAPM+ variants.